Unit 16 – Procedural Programming
Assignment 2 - Design, Build, Test and Evaluate an Application

## Introduction

**Scenario / Vocational Context:** *You are working as a software developer for an events management company. The company has decided that it requires an easy-to-use console-based application that can calculate the cost of having a (big) party. The console application is to be created in Visual Basic, and should be menu-driven, allowing options for users to exit, get help, enter information, and retrieve information.*

*There are a number of potential costs associated. Some are fixed, and cannot be changed, and others are optional (you can choose). You program should display appropriate prompts, menus etc, and retrieve information as required from the user. A suitable results 'screen' should be displayed at the end.*

**Task 1**

## (U16:P3) - Design a procedural application to meet defined requirements

**Task description:** *This task is a DESIGN task, and should be completed before you have begun writing any code.*

Explain the overall purpose of your application

The purpose for the console based application that I will be making is going to be a calculator which will calculate the cost of a party. How I will do this is by making an application which will consist of a menu system which will take the user to questions which have to be answered specifically, either with numerals or text.

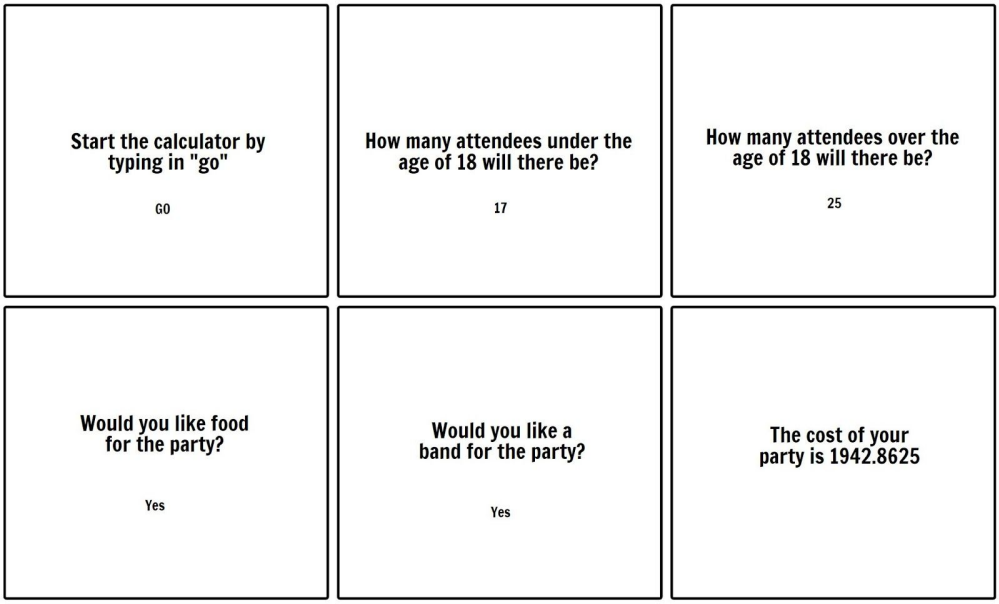
The attendees on the calculator will be calculated based on the user's numerical input, showing how many people are attending the party. The costs of the attendees will vary depending on whether they are under or over 18 years of age. For the over 18s, it will cost £10 per person. For the under 18s, it will cost £7 per person.

The calculator that I will be making will have a pre-added base cost of £500 due to the cost of a venue. The optional costs will consist of a band and food. The band will have a cost of £600 if the user chooses the option of having a band. The food will add 50% for each person that the user has added previously to the calculator.

The application will also feature a main menu system which will have access to three different functions. These functions will be: Starting the party calculator, opening the help screen and exiting the application. I feel that these functions will help the application be a lot more useful to

the user because it gives a much easier interface which can help the user access what is desired.

Provide a ‘worked example’, or story board, of how the application should interact with the user. Provide actual numbers and calculations.




This is a storyboard which I have created to give a rough example of what the application is going to interact to the user.

*Component design:*

*In a table, list the name and datatype of key variables that you intend using.*

| **Variable Name** | **Data Type** | **Purpose** |
|---|---|---|
| userinput | String | The purpose of this variable is so the console can select one of the functions within the menu depending on what the user has typed in. What I mean by this is that the user will type in one of the three options and the console will open the option that the user has typed in. |
| price | Decimal | The purpose of this variable is to collect and gather the price data throughout the calculator which will then be displayed to the user at the end. What I mean by this is that this variable will get all the prices that was created before and add it together. |

| | | |
|---|---|---|
| AttendeesUnder18 | Decimal | The purpose of this variable is to collect the data on how many people will be in the party, which are under the age of 18. |
| AttendeesUnder18price | Decimal | The purpose of this variable is to add data to the price of each person on the party. What I mean by this is that for each person, this variable will add “7” in price. |
| AttendeesOver18 | Decimal | The purpose of this variable is to collect the data on how many people will be in the party, which are over the age of 18. |
| AttendeesOver18price | Decimal | The purpose of this variable is to add data to the price of each person on the party. What I mean by this is that for each person, this variable will add “10” in price. |
| Food | String | The purpose of this variable is to tell the console if the costs for the food in the party should be added or not, which is answered by user input. What I mean by this is the code means whether there will be food in the party or not. |
| Band | String | The purpose of this variable is to tell if the costs for a band in the party should be added or not, which is also answered by user input. In basic terms, this basically means whether there will be a band in the party or not. |

*In a table, list the name and purpose of functions (methods) that you will be using in the application.*

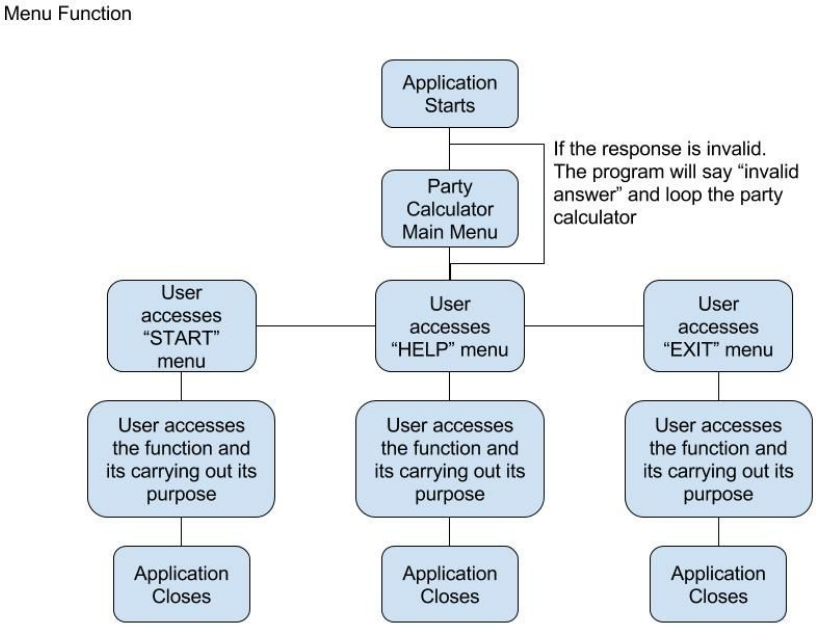
| Function name | Purpose |
|---|---|
| Main Menu | The purpose of this function is that it’s going to be the main menu of the application. This is what the user will first see when he opens up the application, which the user will then see three different options on the application. The options will consist of: “Calculator”, “Help” and “Exit”. The purpose of these functions will be listed below. |
| Party Calculator | The purpose of this function is that it’s going to be the party calculator of the application. This is what the user will see when he selects the “Calculator” option in the main menu, which the user will then see a set of different questions on the application. The questions that will be shown will be questions related to what the user plans on having on the party, so that the budget is calculated. |
| Help | The purpose of this function is that it’s going to be the help menu of the |

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| | application. This is what the user will see when he selects the “Help” option in the main menu, which the user will then see a lot of advice and tips on the application. The tips and advice that will be shown will consist of how to use the program properly and how to overcome common problems if they were to occur. |
| Exit | The purpose of this function is that it’s going to be the exit ability of the application. The sole purpose of this function is so when the user types in “exit” on the application, the application would close automatically. |

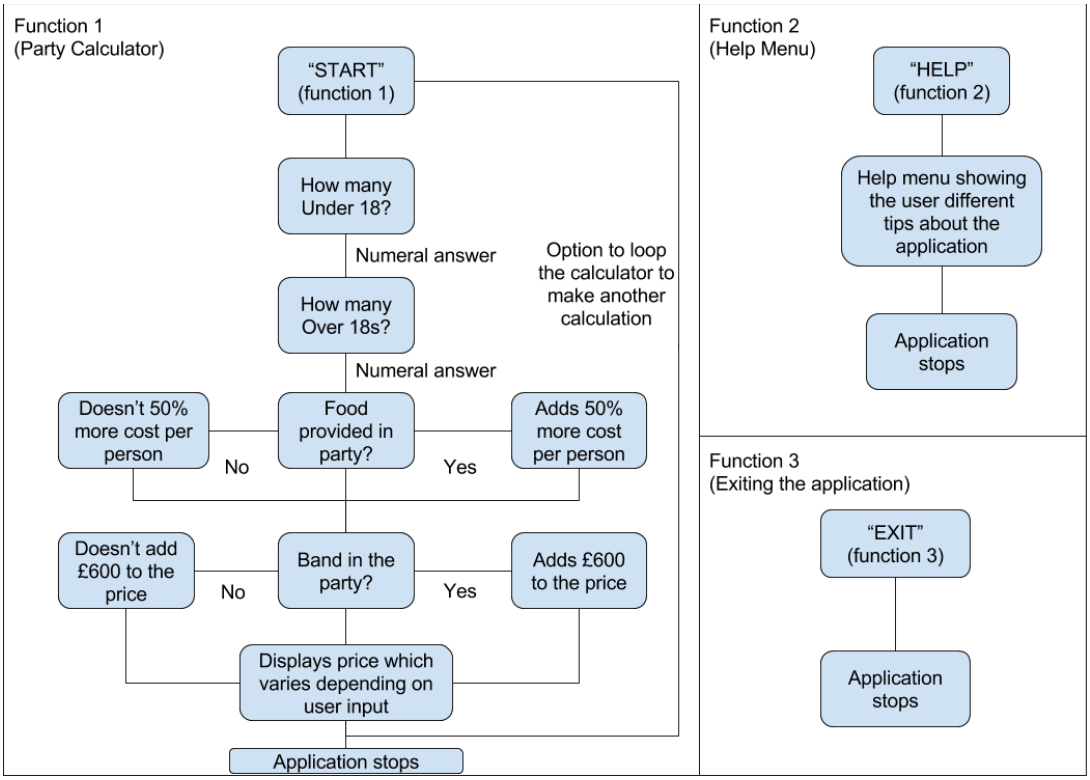
**Task 2**

**(U16:M4) - Create technical documentation for the support and maintenance of a computer program**

Produce flowcharts or pseudocode for two of your methods (functions)




Above is a flowchart of the menu function of the console application.




Above is a flowchart of all 3 functions of the console application.

**Task 3**

**(U16:M2) - Give reasons for the tools and techniques used in the production of a procedural application**

Using examples of at least three different data types in your application, explain why it is important to select the correct data type when creating variables

I have an understanding that, a String variable is basically text, and if we want Visual Basic to store text we just use the word 'String'. String variables are important because this variable is used to collect the text that was input by the user within the application.

This is important because, for the purpose of this application, there are questions that require a yes or no answer. This is where I would use a string variable with the options of "yes" or "no" so the application can use this information to add on to the price or to not do anything to the price.

I also have an understanding that, a Decimal variable is basically numbers that is input by the keyboard, and if we want Visual Basic to store text we just use the word 'Decimal'. Decimal variables are important because this variable is used to collect the the numeral data that was input by the user within the application.

This is important because, for the purpose of this application, there are other question that need quantity to answer it. This is where I would use a decimal variable so the application can use this information to add on to the price.

I learned that Boolean variables is a data type which is used to hold two-state values like true or false, sometimes on or off, and that the default value of the boolean is false, or off. In basic terms, boolean values either means that a specific code is either on or off.

Boolean variables are important because this variable, in my application, is used to open a function such as the main menu, when the user response is either true or false, true opening the main menu and false closing the application. Basically, booleans can be used to access different functions or other forms of code depending on the true or false value that the user or application inputs.

If the wrong variable was used in the application, then the data will not be collected properly and a lot of errors will occur, or it wouldn't make any sense at all. What I mean by this is that each variable has its own situational use, so that it carries data correctly.

For example, if I were to make a variable which will collect text which is user input, I would use a string variable, the reason being is that string variables are made for this type of data. If I were to use a different variable such as Boolean or Decimal variables, the data that the user will input will cause the application to crash or that the application won't be collecting any data at all, making that variable obsolete.

Overall, it's important to select the correct data type when creating variables so that the application carries out its intended purpose without any drawbacks, such as: data not being collected, the application crashing or the application not starting up.

Explain why using the various functions (methods) listed in task 1.3 will make the application easier to read and debug than not having them

Using multiple functions in the code makes the application easier to read and debug because the different types of code that do different purposes are split up in the code. What I mean by this is that the private functions split up the code into different boxes.

This splitting up of the code, makes it a lot easier to read or debug because it is pinned down to specific parts of code. The reason that this makes it easier to read and debug because you can reduce the amount of code that needs to be covered to read or debug if there is an error.

For example, if there was an error within the food variable in the party calculator function, I can read and use the debugger's through the area of the code that contains the food variable within the function to spot any mistakes a lot quicker than if it was a large chunk of code that isn't split up.

Overall, I feel that having various functions are required because it makes the application easier to read because the function splits up different chunks of code that carries out completely different tasks. Which makes it easier to read and pinpoint errors with a debugger because I would know the function that the error is located because the debugger flows with the program.

If my program had no functions it would be a lot harder to find the error within a large chunk of code. This would be very time consuming and frustrating if the error couldn't be found due to mass amount of code surrounding the error.

Explain why some of the various functions (methods) listed in task 1.3 will require input parameters

Various functions require input parameters because this process of storing is defined as name or value pairs. What I mean by this is that input parameters is a type of variable that is used as a subroutine to reference to another piece of data that is provided as input to the subroutine.

Basically, the input parameters are used to contain pieces of data and has the ability to show an existing piece of data, but it depends on the user interaction with the application. Which, in my program will be all the variables such as boolean, string, etc.

For example, if there would be an input parameter implemented within my program, it will contain the pre written price on some data such as the party, band and even the venue, which will then all be added together when the user had added sufficient data and is then shown to the user at the end of the calculator application.

Another example for this is, If i had a base price of 500, due to the venue, and then when an attendee above the age of 18 is called, that value is increased by 10. In this case, the numbers sent to the function is 500 and 10, so this data will be stored within an input parameter, and will write the text “The total price for the party is: 510”.

If the code did not have any input parameters, then it would be impossible to gather data such as the number of attendees, whether there will be a band or food. Which means that the total price cannot be calculated, making the party calculator obsolete.

Overall, I feel that input parameters are important for my program because it can be used to pass a method which is referred to other data, or an “argument”. Which can then the “type” of data that a method can receive, which is what a parameter is referred to, can be helpful in calculating the price or another method which required this type this process due to it holding all the data needed to actually calculate the price.

Explain why return values will be important for the functions (methods) listed in task 1.3

Return values, or return statements, are important because it is where a variable or another type of information which is being returned from the subroutine. What I mean by this is that the return value causes execution to leave the current subroutine and resume at a point of the code immediately after where the subroutine was called, which is known as the return address.

In my program, If i were to use a return value for when I want the user the leave a subroutine that wouldn't be beneficial or have any impact to the calculation or application at all, or if it was the opposite and that the data was extremely important for calculation.

For Example, I could make the user of the application skip over that subroutine due to it not being very important to the calculation itself or I could make the return value carry the data to another subroutine so it can be combined to make a calculation.

Another example would be, if the user were to respond to one of the subroutines, such as “will there be food at the party?”, and the response is a “no” meaning that it is a false response. The return value will be used to leave the current subroutine and resume after this subroutine, because it wouldn’t have any effect on the calculation itself.

However, this method can make code maintenance extremely difficult and error prone. This is why I am implementing at most one or two return statement per function. This is so I can limit the complexity of the code, and that I feel that any more will not be needed in the party calculator since there will be two subroutines which is a string variable.

I feel that return values are important for my program because it helps go over subroutines that are needed within the calculation or if there is a “false” response. However, the “false” response is only applicable for string variables that require a “yes” or “no” answer, which means it's either a true or false. However, a return value isn’t the only method of carrying out the calculation.

For example, I could use “if” and “else if” logic to add or not to include calculations. How I would carry this out is by saying “If Band To.Upper = “yes” Then price = price + 600 Else if Band To.Upper = “no” End If”, instead of using a return value to skip over the subroutine.

Overall, return values can be important to use within a program because of it’s ability to transfer data across. However, it is not the only method of doing carrying this procedure out, explained in the example I provided previously, but it also doesn’t mean it is not required within the program.

This also means that there are a lot of methods on making a program, which also means that there always is a better way of writing code, which is what I will be looking out for on the next version of the program.

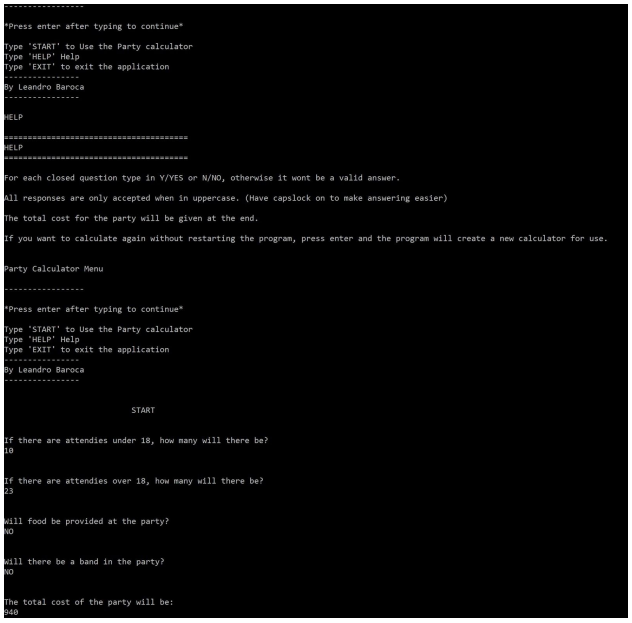
**Task 6**

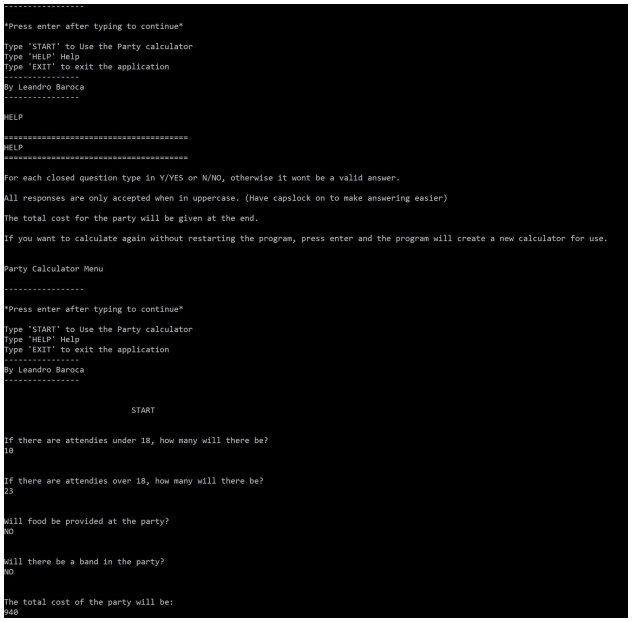
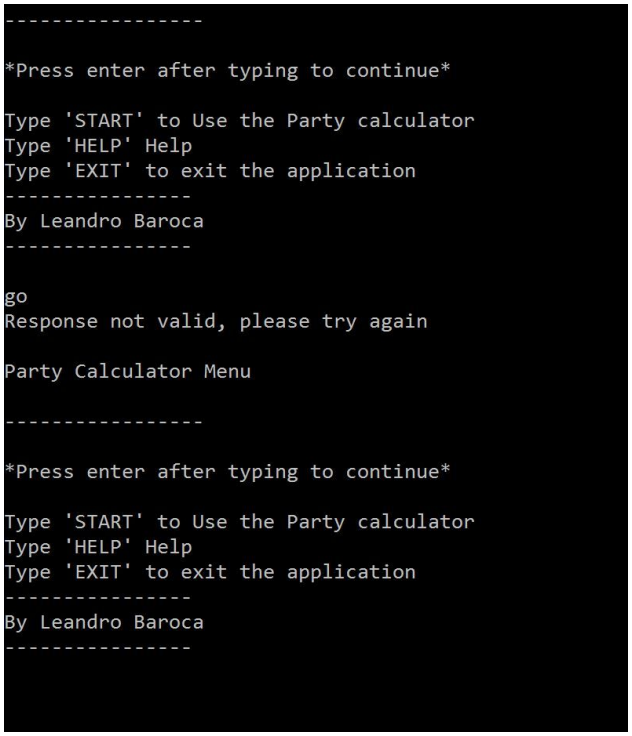
**(U16:P5) - Test a procedural application**

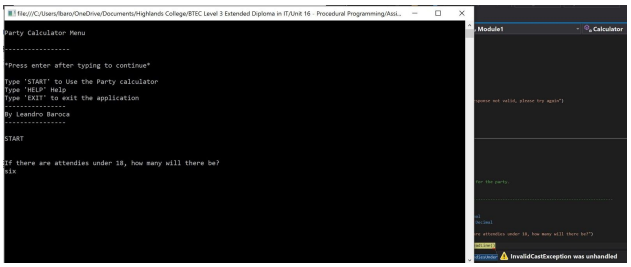
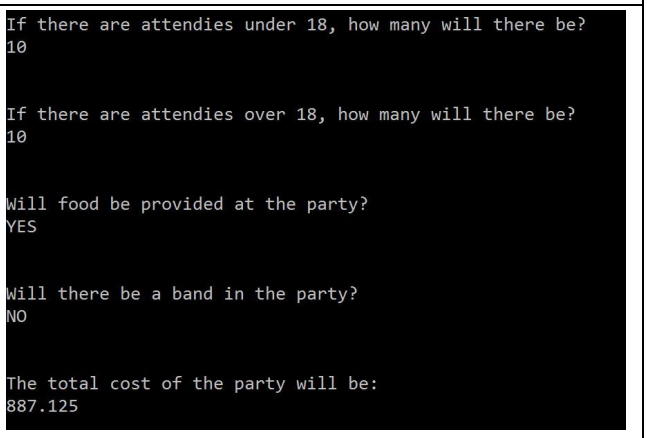
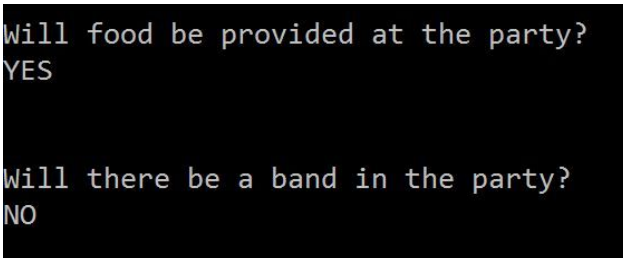
**Task description:** *Test your own program using "black box" tests. To do this, you should: Add tests that test functionality and demonstrate that the program fulfils the design aims and other requirements. This will need at least five (5) tests. You should cover at least one test of each of the following:*

- *Tests to make sure that the key requirements are met.*
- *Tests of correct / expected keyboard input.*
- *Tests of incorrect or unexpected keyboard input.*
- *Tests of calculations.*
- *Tests of 'if' logic.*

*Complete the test plan and provide a log of your testing by producing screen-prints that show the result of each test in the plan. Show these screen shots after the test plan, and number them according to the test number.*

| Test Number | Action | Expected result | Actual Result & Comments |
|---|---|---|---|
| 1. | Test to make sure that the key requirements are met.<br><br>To test for this, the action I will be taking is seeing if the program is related to calculating the price of a party as much as possible. | The expected result for this test is that the application will have the key requirements, which it has to be menu-driven, allowing options for users to exit, get help, enter information, and retrieve information. |  <br>With the test results, I saw that the application meets the requirements of it being menu-driven, allowing options for users to exit, get help, enter information, and retrieve information. Therefore, the application does carry out its main purpose. |

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| 2. | Test of correct / expected keyboard input.<br><br>To test for this, I will be typing in the correct keyboard input in the console application. | The expected result for this test is that the application would run smoothly without crashing with the correct input. |  <br>With the test that I carried out above, I could also see that the application works smoothly with the correct keyboard input. This means that the application doesn't crash when the user inputs with the keyboard and that it meets the expectations. |
| 3. | Test of incorrect or unexpected keyboard input.<br><br>To test for this,I will be typing in the incorrect keyboard input in the console application. | The expected result for this test is that the application would respond with "invalid answer" and not crash due to the incorrect input. |  <br>When I carried out the test on the main menu, the application responded with a "response not valid" error message and looping the menu. |

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| | | | The main menu of the application meets the expected result due to the application not crashing.<br>  <br><br>However, when I tested on the calculator itself, the application didn't respond with a "response not valid" error message and crashed a few seconds after typing the incorrect input. Therefore the calculator itself does not meet the expected result |
| 4. | Test of calculations.<br><br>To test for this, the action I will be taking is using the application for its intended purpose and find out whether the calculations are accurate. | The expected result for this test is that the price that is shown at the end of the calculator will be the correct price with the appropriate choices and other additions in price. |  <br><br>The actual result of the test met the expected result because the calculation that is shown in the end is the correct calculation for the selected data. Therefore the program does carry out calculations correctly. |
| 5. | Tests of 'if' logic.<br><br>To test for this, the action I will be taking is the "IF" statements that I have created in the program, which is the responses that the string | The expected result for this test is that each "if" or "else if" logic will work properly within the application without it crashing or showing "response invalid". |  <br><br>From the test above, I could also test the "if" logic within the party calculator to check if it crashed on the parts that contained "if" or "else if" logic. The result above shows that the |

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| | variables will accept. | | program still continued after typing in the data so it carries out the expected operation. So no further action is needed to correct it. |

**Task 7**

**(U16:M3) - Analyse actual test results against expected results to identify discrepancies**

**Task description:** *Analyse actual test results against expected results to identify discrepancies for two tests that failed:*

Explain the difference between the actual and expected results and why this difference came about

While I was creating the program, I have come across many errors by using the debugging tools and errors within my code. Most, if not all, of these bugs consisted of misspelling of variables or the wrong datatype for the wrong variable. I found that these errors were easily fixed because they were minor misspellings.

However, there are some errors that I have found within the test due to the fact that there wasn't a code implemented for that specific area. The one error that I have noticed so far with the testing is that there is no code that prevents the program from crashing if the user was to type in an invalid answer to the questions of the application.

For example, when I launch the application and select the party calculator, then I type in a response which is an invalid response, the application freezes then crashes. I feel that this is an issue that needs to be addressed on my next version of the program because this can be very problematic. Due to the reason that the user will have to redo everything if the program was to crash, which can be very frustrating to use.

Overall, I feel that this error is an annoyance to the user due to the program crashing, even if there was a very minor mistake or mistype. However, this error can be fixed with an addition to code. Which is why I am planning to add code to my next version of the program to address this issue of the program crashing.

Explain how you would correct the problem in the next version of the program

I could correct this problem in my next version of my program by making code similar to the one I made on the main menu, which gives a message saying “invalid answer” if the user was to type in an invalid answer within the question instead of the program crashing.

I already have a code implemented within the application that prevents it from crashing on the main menu of the application, but it is only working for the main menu itself. This means that I will need to upgrade the code on the calculator itself in my next version.

For instance, the code consist of “Case Else Console.Writeline(“Response not valid, please try again”). This means that I already have a reference code to use when making a crash proof code on the party calculator, which is beneficial for the upgrade because I have proof of code that works correctly and prevents the program from crashing.

Overall, I feel that this problem is quite an important error which can be addressed easily because I already have code to reference from which I know works due to the testing that I have carried out.

If I were to implement this code into the next version of the application, I believe that it would make the application much more robust due to it not crashing if the user has incorrect keyboard input. Meaning that it would reduce a lot of annoyance and overall make the application much more reliable to use, without worrying for the risk of it crashing.

**Task 8**

**(U16:D2) - Evaluate a procedural application**

**Task description:** *Evaluate a procedural application. Evaluate your program by discussing:*

<u>How well it meets the aim of the application set out in the design specification</u>

I feel that the application that I have created meets the intended purpose that I have stated in the beginning well. The reason for this is because the application gives an accurate calculation on what the user would want to calculate for the party.

In the design specification, it was specifically said that the application has to be menu-driven, should allowing options for users to exit, should allow the user to get help, allows the user to enter information and retrieve information.

With the options that allow the user to exit, I implemented a code in the main menu which tells the user if the words “EXIT” are typed out into the application and then enter is clicked, the application will close automatically.

With the allowance for user to get help, I implemented a code in the main menu which tells the user when the words “HELP” are typed out into the application and then enter is clicked, the application will write out text which was pre-written in the code by myself.

The text that I wrote myself gives advice and tips to the user on how to use the application much more easier and with other common problems I have encountered myself. I felt that this would make the experience easier to the user if common problems are faced when using this application.

With allowance for the user to enter information, it is a very simple process of the user typing in the answer to the question that appears in the program. However, it has to be very specific answers. Which is why I have tips on the Help function that tells the user the important things to do on the program and how to answer the questions correctly without the program being bugged.

With allowance for the user to retrieve information, I believe that it is also a very simple process because the application shows, what I feel, that is the only necessary amount of information for a party calculator. For instance, after answering all the questions, the user gets the final calculation at the end of the calculator, which I feel is the only reason that this application was made.

Also, the help function gives advice and tips for answering questions properly and how to avoid common problems I found, however, I feel that the help function could contain more data which helps address possible problems which I have not found.

Overall, I feel that the application that I created does carry out its intended purpose that I have set in the design specification. However, I feel that this application can be vastly improved due to its very simple nature. Improving aspects such as crash proof code, making it more aesthetically pleasing and other improvements that make it much more user friendly would make the application a lot easier to use for the user.

Where it does not meet the requirements well (or at all) and what could be done to resolve this

I feel that the application doesn't meet the requirements well in specific areas. One of these areas is the possibility for the user to exit the application when the user is in a specific stage of the application.

For example, if the user is using the calculator and then the price is given. It would be a better system if there was an option to either restart the calculator, send to the main menu or to exit the application. However, the code that I implemented only lets the calculator loop, meaning that it makes a new one after completion, until the user closes the program manually.

With this requirement, I feel that it should be required within the program because it makes it seem like I didn't put much effort for the user with the code. So having this type of "requirement" in my application would make it much more user friendly and a much better application in general.

Another area which I feel that the requirements could be met better is the help function that I have implemented in my program. The reason for this is due to the fact that I believe that the information on the help function is very small and that it only covers common errors I found and how to write out a question properly.

What I mean by this is that I feel that the help function doesn't say enough information to help the user overcome problems which are possible that I have not yet found, therefore making the user confused on what to do to overcome the problem and leaves the application without using it.

Another requirement is to have a much better help system. I also feel that it should be required within the program because it is very risky to have very little help, which can leave the user stuck. Also having this type of "requirement" in my application would make it less risky to use and yet again more user friendly.

The final requirement that I feel should be implemented that I have not mentioned before, is including the price with and without the VAT rate, especially in a way which the user can read it. I feel that this would be useful so the user can see the difference between the price with and without VAT, and figure out how much extra the user will need to pay towards the party itself.

For instance, if the user has a party with 20 under 18s and 36 over 18, without food and without a band, the party will have a total cost of £1000 without VAT and 1175 with VAT. This would make the program a lot better to use if the user wants an accurate budget cost with the VAT in place, so that there is no surprise expenses.

Overall, I feel that these requirements are important to include in my program because it would make it a much better calculator and program to use in general. If I were to install these requirements within my application, it would be a much more user friendly and accurate calculator due to it having a lot more accessibility, more options to help and more accurate calculations so that the user has the right budget when making a party.

Coding in your program that you are particularly pleased with

The coding that I am very pleased with is the code that required to add the prices together and multiplying them with the required user response. I am pleased with this because I found it difficult to use the mathematical aspect of the program since I am a very inexperienced programmer.

I am pleased by this coding because it was very difficult to do but I managed to accomplish it at the end and turned out to work properly after some testing of the calculations, which made it a very satisfying completion. It was also very satisfying when I completed it because I initially thought that I couldn't get it completed alone.

Another code that I was pleased with was the code to create the main menu system. I am pleased by this because I was unsure of how to make a menu system since I don't have much experience in doing so, but I managed to get it complete and working properly which also surprised myself because I did not expect it to work and that I felt accomplished when I finished the code with no errors.

I am very pleased that the menu system worked at the end because it makes it a much more accessible application because of the new help option and exit the application option for the user to access with ease. When I started making the menu code, I initially thought that it would be almost impossible for me to do but I completed it with a little help from research from areas which I didn't understand.

Overall, I am pleased with my whole program, but these types of code are the ones I was most pleased with because of how difficult they were for me due to my inexperience with these types of code. Overall, I feel that these also were the most important code out of the rest because without these, the calculator wouldn't do its purpose of being a party calculator. However, without the rest of the code the program wouldn't work in general so the rest is still very important.

Coding in your program that you found difficult or think could be improved

The coding I found very difficult is the mathematical code to do with pricing, which is same code that I am pleased with, due to the reason that I am a very inexperienced programmer with very basic knowledge about programming, and that this was code that I haven't looked much into which made it a lot more difficult that it should have been.

With my time throughout Visual Basic, I didn't have a lot of experience with mathematical code. Which is why I found this type of code very difficult to do. To complete this code within the program I had to do some research on how to do the mathematical aspect of the coding, so that I can make the application do it's intended purpose of calculating.

The code that I think should be improved within the program is to implement a code within the party calculator which stops the program from crashing when the user has an incorrect or unexpected input. As I also mentioned before, from the testing there has been an error which I found in which there is a lack of code that makes the calculator itself crash proof.

What I mean by this is that the application just crashes if the user gives an unexpected keyboard, causing the application to be restarted. I can imagine this being a mass annoyance for a user that has common typing mistakes or that didn't follow the instructions in the help menu.

Another improvement that I think should be implemented into my code is the ability to either send the program back to the main menu or to restart the calculator when the user has finished

with the calculator. I have also stated this problem before, but this is also another concern I have that should be improved within the program.

The reason for this being that it gives a lot more options for the user when using the application, instead of being restricted to what the application is scripted to do and just follow along. I feel that this code is important to implement since it makes the user experience a lot better when using the application.

Yet another improvement that I would personally want is to make the application is to implement better help for the user since there is very little information on the help function. I already have stated this many times before, but I feel that this should be improved since it makes the application a risky program to run if there is barely any help to aid the user on what to do when something doesn't go according to plan.

As I have previously mentioned, having a better help system should be required within the program because it is very risky to have very little help, which can leave the user stuck, because this would make it less risky to use and more user friendly.

The final improvement that I feel that should be added is more clear instructions throughout the application. I haven't mentioned this before due to the recent realisation that the user might be very confused when using an unknown application.

For example, instead of just having comments within the code itself, I should implement a key system to show how to access some parts of the application to a user that has no idea on how to use the application.

Conclusion

In conclusion, I believe that my program has met its initial purpose well, which is to calculate the costs of a party depending on user input properly. However, I feel that there is plenty of room for improvement in my application.

With Visual Basic being a simple programming language, it gives it the ability to have almost limitless customisation, with the biggest limit being that it's console based. This means that I can tailor the program to the user's needs and further improve the application so it is more aesthetically pleasing and easier to use without the need of help.

Overall, I feel that my application is at a very good stage due to its functionality and coding. However, it is nowhere near perfect. Which is why I want to bring the program as close to perfect as I can in the next versions.